A REVIEW ON: SPIKING NEURAL NETWORKS NEED HIGH-FREQUENCY INFORMATION

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ABSTRACT

This report follows from the methods, results, and conclusions presented in the paper "Spiking Neural Networks Need High-Frequency Information". We validated the paper's core claims as full MaxFormer reached a top-1 accuracy of 96.64% on CIFAR-10 and 81.44% on CIFAR-100. We extend the model with learnable-β and beat the baseline by 0.49 percentage points. Comparative analysis of token-mixing operators revealed that high-frequency amplifiers like LearnedHP and Laplacian filters outperform max-pooling, supporting the theory that SNNs benefit from active high-frequency restoration. Temporal analysis on the SHD dataset confirmed that while max-pooling restores high-frequency energy along the cochlear axis, it does not recover temporal high-frequency dynamics suppressed by LIF neurons. Code is available at: `https://github.com/COMP6258-Reproducibility-Challenge/SNNs_need_HF_info`

1 BACKGROUND

Spiking Neural Networks (SNNs) transmit information through binary spikes, offering energy-efficient computation but historically trailing Artificial Neural Networks (ANNs) in accuracy. The conventional explanation is *representation error*: the claim that binary activations inherently limit feature precision. Fang et al. (2025) reject this account and instead show that the LIF charging equation acts as a first-order IIR low-pass filter, with decay factor β controlling the cutoff, and the effect compounding across L layers. This manifests as a +2.39% gain on CIFAR-100 from swapping Avg-Pool for Max-Pool, motivating **MaxFormer** (Fang et al., 2025). SNNs are of growing interest for deployment on neuromorphic hardware such as Intel Loihi and IBM TrueNorth, where binary spike communication enables lower energy per spike costs. Closing the accuracy gap with ANNs is therefore an active research priority, and understanding why that gap exists is as important as narrowing it.

The first motivating gap in the paper is the choice of $\beta = 0.5$, treated as fixed, despite the transfer function predicting it as the key tunable; we ask whether a swept or learned β does better. Second, the high-frequency claim is supported only by contrasting Max-Pool with Avg-Pool, leaving open whether the benefit comes from high-frequency preservation specifically or from max-pooling as an operator; we test this with min-pool, Laplacian, and learned high-pass alternatives. Third, the low-pass theory is temporal (over simulation timesteps) but is validated only on spatial image features; we extend the analysis to the temporal axis using the Spiking Heidelberg Digits (SHD) dataset.

2 ESTABLISHING A BASELINE

We reproduce experiments from Fang et al. (2025) using the authors' released codebase and configuration files. We train two lightweight SNNs, AvgFormer using average-pooling, and MaxFormer (max-pooling) on CIFAR-100. The second reproduction trains the full MaxFormer architecture on both CIFAR-10 and CIFAR-100. Walltime limits on Iridis prevented full 400-epoch training, but both Full MaxFormer runs plateaued before termination. Table 1 confirms the paper's claim in direction and ordering, serving as a baseline for our extensions. All extensions (learnable-β, alternative operators, SHD analysis) were implemented by us on top of the authors' released codebase.

Table 1: Reproduction results. Δ top-1 accuracy minus the paper's reported value (percentage points).

Model	**Dataset**	**Paper (%)**	**Ours (%)**	Δ **(pp)**	Epochs
AvgFormer (SNN, avg-pool)	CIFAR-100	76.73	73.82	-2.91	264
MaxFormer-lite (SNN, max-pool)	CIFAR-100	79.12	76.38	-2.74	264
Full MaxFormer (SNN)	CIFAR-10	97.04	96.64	-0.40	336
Full MaxFormer (SNN)	CIFAR-100	82.65	81.44	-1.21	336

3 β Sweep

We performed four runs of MaxFormer at $\beta \in \{0.2, 0.5, 0.75, 0.9\}$ for 200 epochs each, the hyper-parameter setup matches that of the original paper's Table 6; Table 2 shows the accuracies achieved.

Table 2: Fixed-β sweep and learnable-β run on MaxFormer, CIFAR-100.

β	Best Top-1 (%)
0.20	78.02
0.50	78.86
0.75	77.72
0.90	73.52
Learnable (per-layer)	**79.35**

We replaced every fixed LIF neuron in MaxFormer (28 instances) and Spikformer (Zhou et al., 2022) (37 instances) with a parametric LIF whose decay β is a learnable scalar per layer, parameterised as $\tau = 1 + \text{softplus}(\theta)$ and initialised at $\beta = 0.5$. As shown in Figure 1, 22 of 28 layers reduced β below the default, with the largest reductions at the attention $Q/K/V$ projections. The network shifts β in the direction the sweep shows is optimal, confirming the paper's choice of $\beta = 0.5$ is defensible, while learnable β yields a further $+0.49$ pp gain.

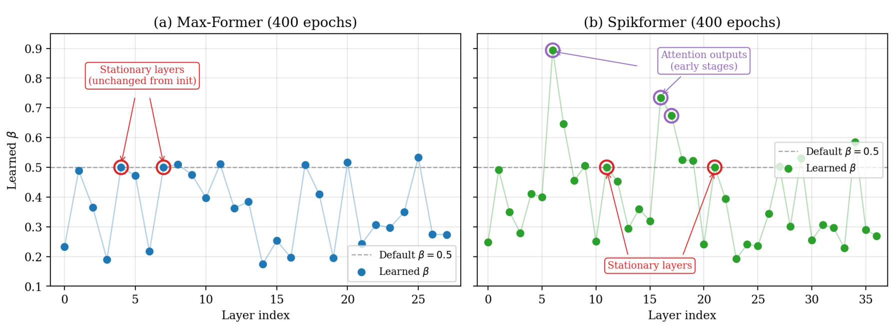

Figure 1: Learned per-layer β in Max-Former after 400 epochs.

4 High-Frequency Operators

To investigate whether max-pooling specifically helps because it preserves high-frequency content, or whether simply swapping out average-pooling for any other operator would yield similar gains, we trained three additional variants of MaxFormer-lite on CIFAR-100, each replacing the pooling layer with a different operation. This involved implementing min-pooling, a fixed Laplacian filter, and a learned high-pass convolution.

Table 3: Alternative token-mixing operators on CIFAR-100, 264 epochs.

Operator	Best Top-1 (%)	Δ vs AvgFormer (pp)	Δ vs MaxFormer-lite (pp)
AvgFormer (baseline)	74.59	—	-1.79
MinPool	74.94	$+0.35$	-1.44
MaxFormer-lite	76.38	$+1.79$	—
Laplacian	77.27	$+2.68$	$+0.89$
LearnedHP	**77.86**	$+3.27$	$+1.48$

All high-frequency token-mixing operators outperform AvgFormer, while MinPool gives only a small gain. LearnedHP and Laplacian also outperform MaxFormer-lite by $+1.48$ pp and $+0.89$ pp respectively. The ranking LearnedHP $>$ Laplacian $>$ MaxFormer-lite $>$ MinPool $>$ AvgFormer follows how directly each operator targets high spatial frequencies, suggesting that the benefit is not max-pooling specifically, but high-frequency restoration more generally.

5 Temporal Analysis

5.1 Experimental Design

To investigate whether the claims of Fang et al. (2025) extend beyond the spatial domain, we designed a temporal extension using the Spiking Heidelberg Digits (SHD) dataset (Cramer et al., 2022). SHD requires no analogue-to-spike conversion, spike timing is essential for good performance (Cramer et al., 2022), and its 700 cochlear-ordered input channels provide a natural spectral structure for testing high-frequency suppression.

Two-Axis Analysis. The paper's low-pass derivation is temporal, operating over the discrete timestep axis T, yet is validated only through spatial frequency analysis of 2D image feature maps (Fang et al., 2025). We therefore compute the 1D DFT of intermediate activations separately over (i) the cochlear channel dimension L, asking whether max-pooling restores suppressed high auditory-frequency content, and (ii) the timestep dimension T, asking whether temporal high-frequency attenuation predicted by the LIF transfer function is observable and SNN-specific.

Models and Training. We trained four models on SHD, mirroring the ablation in Fang et al. (2025): **SNN-Avg**, **SNN-Max**, **ANN-Avg**, and **ANN-Max** (ReLU replacing LIF), each across 5 random seeds. All hyperparameters follow the neuromorphic configuration of the original paper: AdamW, cosine schedule, lr $5{\times}10^{-3}$, weight decay 0.06, 96 epochs, batch size 16, mixup, $\alpha{=}0.5$. We extract activations at four stages via forward hooks and report the high-frequency energy ratio (spectral energy above normalised frequency 0.3).

5.2 Results

Table 4 shows SNN-Max outperforms SNN-Avg by $+0.37\%$ (93.12% vs. 92.75%), consistent in direction with the paper's $+2.39\%$ on CIFAR-100, with the smaller gap expected on a simpler task. ANN-Max scores -0.24% below ANN-Avg, suggesting max-pooling gives no benefit without LIF neurons, which mirrors the paper's claim that the effect is SNN-specific. ANN accuracy is substantially lower overall, likely due to gradient instability when replacing LIF with ReLU in attention layers, so this comparison should be interpreted with caution.

On the cochlear axis (Figure 2, left), SNN-Max retains more high-frequency energy than SNN-Avg at the final layer (HF ratio 0.122 vs. 0.093), replicating the spatial finding of Fang et al. (2025). The temporal axis (Figure 2, right) tells a different story: SNN-Max does not recover temporal high-frequency energy (0.270 vs. 0.276), while both ANN conditions retain more than either SNN condition (0.322, 0.325). This is consistent with the LIF transfer function prediction, and expected since our max-pooling operates over L, not T.

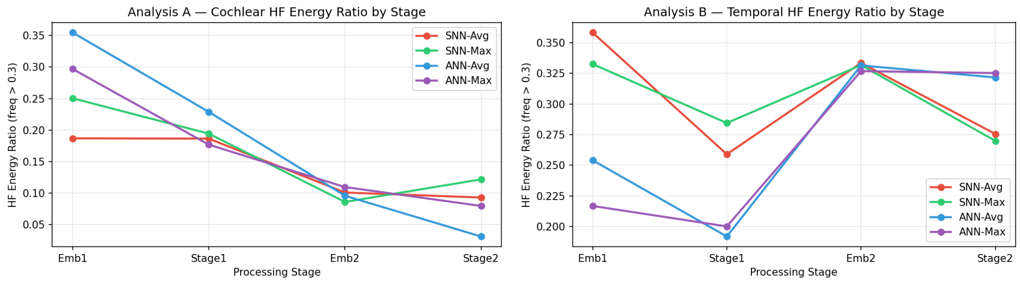

Figure 2: High-frequency energy ratio across processing stages. *Left:* cochlear axis. *Right:* temporal axis. Solid: SNN; dashed: ANN.

Table 4: Mean top-1 accuracy on SHD across 5 seeds.

Model	Mean Acc. (%)	Std	Best
SNN-Avg (low-pass baseline)	92.75	0.15	93.02
SNN-Max (high-freq. token mix)	93.12	0.29	93.55
ANN-Avg (ReLU, low-pass)	71.93	0.48	72.61
ANN-Max (ReLU, high-freq.)	71.70	0.24	71.91

6 Conclusion

We reproduce the main direction of Fang et al. (2025), with Full MaxFormer reaching 96.64% on CIFAR-10 and 81.44% on CIFAR-100. Overall, our results support the paper's claim that the ANN–SNN gap cannot be explained purely by binary spikes, and that frequency suppression caused by LIF dynamics is important.

However, our extensions show that the story is more nuanced. Learnable-β gives a small improvement, while LearnedHP and Laplacian operators outperform max-pooling despite using different mechanisms. This suggests that the original gain is not specific to max-pooling itself, but to the broader principle of preserving or restoring high-frequency information.

Our SHD experiments further complicate the original interpretation. Max-pooling restores high-frequency energy along the cochlear axis, but not along the timestep axis. This matters because the paper's theoretical motivation is temporal, yet the architectural solution mainly acts spatially. Therefore, our results partially validate the low-pass filtering theory while showing that spatial high-frequency restoration alone is not enough to overcome temporal information suppression in SNNs.

Limitations and Future Work. Walltime constraints prevented full 400-epoch training, and SHD ANN baselines remained weak. Future work should test operators acting directly along the timestep axis, such as temporal depthwise convolutions or learnable temporal high-pass filters.

References

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